European Developer Conference



New Platform Technology

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Introduction

A key factor in the success of the Palm OS platform is the diversity of devices offered by our licensees.

Staying competitive means leveraging their innovation in your applications.



Agenda

Looking Better

- QVGA
- Dynamic Input Area

Sounding Better

- Sampled sounds

Running Better

- Faster, slower, cleaner

Looking Better

Quarter VGA Screens

Dynamic Input Area Support

Quarter VGA (QVGA)

1.5x density screen

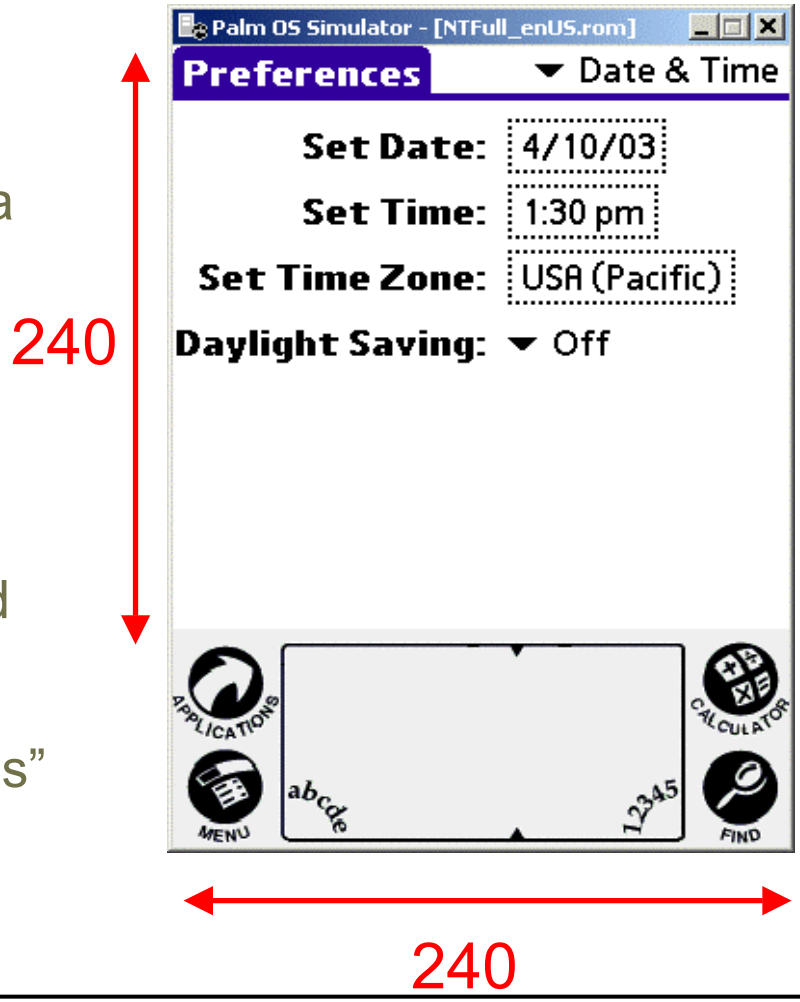
- 240 x 320 including input area
- Active input with inking
- Maybe collapsible

Much like 2x density

- Bitmap and font families
- 160 x 160 standard coord grid

KB article 495

- “Optimizing for QVGA Screens”



QVGA Scaling

1.5 x odd number = rounding error

- Some things rounded up, some rounded down
- This does get complex, but few apps are affected

Every other pixel doubled for bitmaps

- Size clipped to fit “intended size”

No 2x ◄► 1.5x bitmap scaling

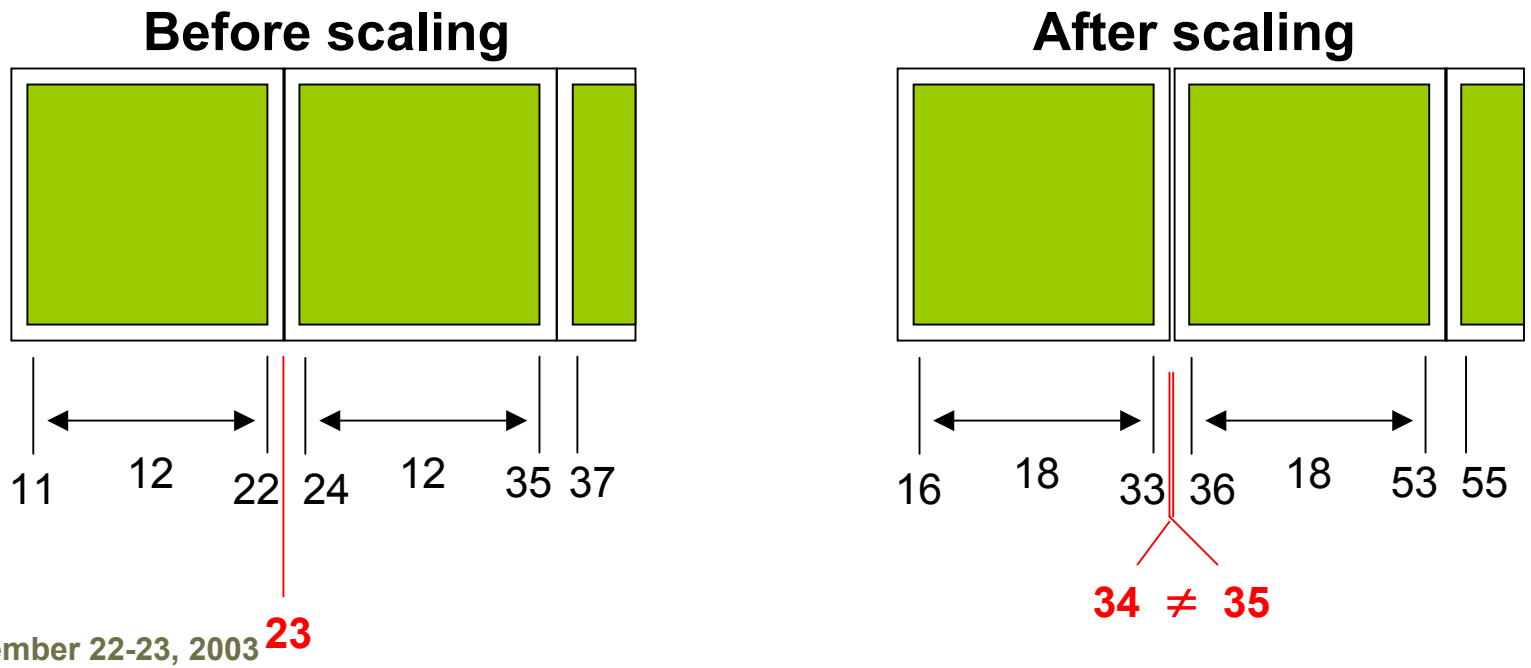
Palm OS 5.3 Simulator has QVGA mode

- Set in display settings
- Even though OS 5.3 is Chinese there is English version of Simulator

Push Button Example

Rectangle frames are drawn around UI objects bounds

- Use odd extent values for push buttons to avoid this



QVGA Quick Tips

Generally use even values for topLeft coords

Add 1.5x members to bitmap families

- In addition to single density

Be careful when form objects are close together

Use native coordinates for custom drawing

- Never leave OS in native coordinates

Use `nativeFormat` offscreen windows

Use `WinSetScalingMode()` to prevent bitmap scaling

Animation: move in even pixel amounts

Dynamic Input Area (DIA)

2 New Features

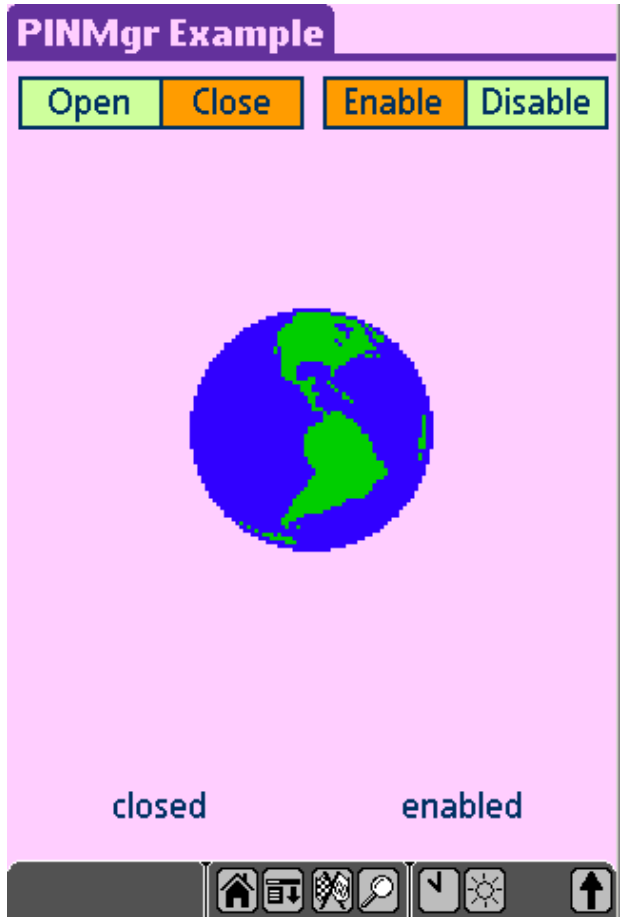
- Inking
- Collapsible

Existence of PinMgr doesn't mean the input area is collapsible

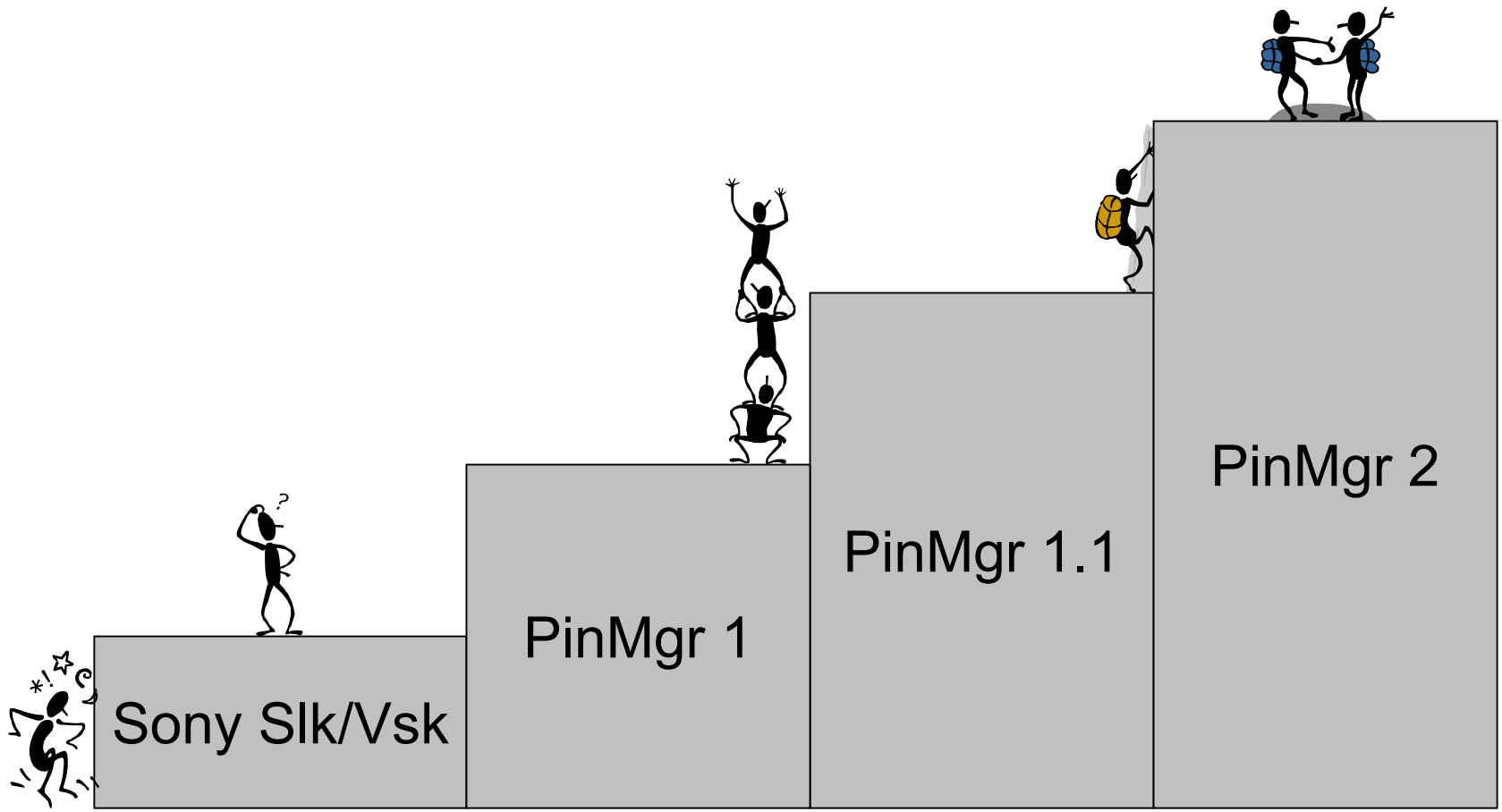
- Just that it is safe to use the calls

Article 507 in Knowledge Base

- "Using Palm OS PInMgr and cross platform DIA support"



Variations on a Theme



PinMgr and Sony Vsk

Main differences

- For Sony API enable on app by app basis
- PinMgr enable form by form at frmOpenEvent
- Sony sends different notification at resize

Things in common

- Method to resize form and move objects

Sample code in Article 507 abstracts both APIs to `Collapse…` calls that work across different devices

- Sony, Palm Inc, Garmin, TapWave, OS 6
- Portrait and Landscape
- Most apps can just use these utility calls and work
- Other apps can modify utility code

Collapse - Set up and Clean up

At `AppStart` call `CollapseAppStart`

- Inits version of API
- Registers for notifications
- Opens Sony library if on Sony

At `AppStop` call `CollapseAppStop`

- Unregisters for notifications
- Restores state on older Sony API

At `frmOpenEvent` call `CollapseSetState`

- Boiler plate form set up for PinMgr API
- Sets initial input area state for form

Collapse - Reacting to a resize

At `frmOpenEvent` and `winDisplayChangedEvent` call `CollapseResizeForm()`

- Tries to resize form or dialog to currently available extent
- Returns true if resize needed and amount of horizontal and vertical resize offsets

If it returns true move form objects with `CollapseMoveResizeFormObject()`

- Object movement/resizing according to what makes sense for the form
- Use returned horizontal and vertical offsets as basis for movement and resizing
- Text fields: `FldRecalculateField()`

Post `winDisplayChangedEvent` in response to `winEnterEvent`

- In case display area changed outside of your event loop (alerts, etc.)

Sounding Better

Checking for Sound

SndPlayResource()

SndStreamCreate()

Streaming from a VFS card

Checking for Sound

Check for Palm OS sound feature

- For Tungsten T, also check device ID

```
FtrGet( sysFtrCreator, sysFtrNumOEMCompanyID, &Manufacturer );
FtrGet( sysFtrCreator, sysFtrNumOEMDeviceID, &DeviceID );
if (( Manufacturer == (UInt32)('Palm') && DeviceID == (UInt32)(
        'Frg1' )))
```

Sony Sound

- Get Yamaha ADPCM chip library from Sony
- Moved to our API with OS 5.2 products

SndPlayResource()

Check to see if Sound Stream Feature Set is present

Supported WAVE parameters

- Uncompressed PCM
- IMA 4-bit ADPCM (aka DVI ADPCM, or WAVE format 0x11)
- One or two channels
- All normal sampling rates (8k, 11k, 22.05k, 44.1k, 48, 96k)

API does not exist on OS 5.0 Sony Products

- Convenience API but limited

SndStreamCreate()

Requires Sound Stream Feature Set presence

Creates audio data stream

- Record or playback uncompressed sampled audio data
- Number of streams: 1 input, up to 15 output

Create small number of streams on app init

- Blit PCM data onto stream buffer as needed

Streaming Sound from a VFS Card

Sample code in Knowledge Base article 40

- Reads raw PCM data from expansion card

Sample pipes entire PCM file through the Sound Manager

- Including header, which doesn't sound pretty
- Remove header from PCM

Requires Palm OS 5 SDK

palmsource

Running Better

Faster

Slower

Cleaner

Running Faster

Palm OS 5 system routines are ARM native

Use PACE Native Objects (PNO) to improve performance

- Compression, encryption, etc.
- No advantage if primarily calling system APIs
- Code, measure, code again...

Running Slower

Some apps may need less speed

- User perception / reaction time
- Hyper-speed scrollbars
- Graphics
- Getting pen events

`SysTaskDelay()` is your friend!

Running Cleaner

Test with Gremlins on Emulator with Debug ROMs

Test with the Simulator

Check for feature sets

Read the Compatibility Guide

- *Palm OS Reference*, Appendix

Running Cleaner – Common Problems

Shadow structures

- Misuse is #1 cause for Palm OS 5 incompatibilities

System-use only functions

- *Palm OS Reference*, Appendix

Graffiti 2

- Changes effects small number of apps

High Density

Running Cleaner - High Density

Call WinSetCoordinateSystem() before drawing

- System resets to standard coords for UI drawing

Bitmaps

- Always drawn at appropriate density
- Use bitmap families
- Use offscreen trick to shrink low density bitmap

Knowledge Base article 341 “High Density Overview”

- With sample code

Q & A